OpenAI Normalize Cost Budget

Based on live production data for Danish (user id 4) · Generated 11 Nov 2026 · Model: gpt-4o-mini · PKR rate used ≈ 280

Bottom line: One unique message that gets normalized ≈ **$0.0004** (~0.10 PKR). A user with Danish-like volume across **4 groups** costs roughly **$1.50–$2.50 / month** in OpenAI — far below the PKR 7,000 plan.

1. What we bill OpenAI for

Each scraped WhatsApp message that needs structuring makes **one** `gpt-4o-mini` Chat Completions call. Same body + same footer already normalized → **OpenAI is skipped** (dup-skip). Short / empty messages also skip the model.

Hits are logged in `openai_api_hits`. Danish currently accounts for essentially all recorded hits (~4,290).

2. Cost of a single API hit

Token mix for a typical Danish normalize call (chars÷4 approximation):

Component	≈ Tokens	Notes
System prompt + JSON schema	~2,570	Identical every call → prompt-cacheable
User message (avg / median)	~100 / ~54	Danish avg message ≈ 398 chars
Model output JSON (est.)	~250	Listings payload; varies by multi-offer msgs

List prices (OpenAI, Nov 2026): input **$0.15** / 1M · output **$0.60** / 1M · cached input **$0.075** / 1M.

Scenario	≈ USD / hit	≈ PKR / hit	≈ USD / 1,000 hits
With prompt caching (realistic)	$0.00036	0.10 PKR	$0.36
No cache (conservative)	$0.00055	0.15 PKR	$0.55
Budget rule of thumb	$0.00040	0.11 PKR	$0.40

Use **~$0.0004 per unique normalize** for forward planning unless you disable caching or change models.

3. Danish — unique msgs normalized per group per day

Danish has **6 monitored groups**. Counts below are distinct content fingerprints that produced at least one `normalized_messages` row.

Window	Avg / group / day	Median	All his groups / day
All history (~54 active days)	40.3	34	~164
Last 7 days	56.0	43	~336

Monitored groups (lifetime unique-norm avg / day)

Group	Letter	≈ Avg unique / day
DHA Inventories	e	67.1
A-1 Estate Builder & developers	c	52.6

Group	Letter	≈ Avg unique / day
BTK Fresh inventory	f	48.5
DHA KARACHI INVENTORY HUB	b	34.2
AMAL ASSOCIATES	d	29.1
DHA Karachi | Realtors	a	24.0

Actual OpenAI hits (last 7 days): **~251 / day** — about 75% of unique norms; the rest are dup-skips / short messages. Lifetime hit average while logging: **~286 / day**.

4. Budget formula — average user with 4 groups

If a user has **X** unique messages per group per day that need normalize, and monitors **4** groups:

```
hits/day ≈ X × 4
cost/day ≈ X × 4 × $0.0004
cost/mo ≈ X × 4 × $0.0004 × 30
```

X (unique / group / day)	Hits / day (4 groups)	≈ $ / day	≈ $ / month
34 (Danish median)	136	$0.05	$1.50
40 (Danish lifetime avg)	160	$0.06	$1.70
56 (Danish recent 7d)	224	$0.08	$2.40
100	400	$0.16	$4.80
200	800	$0.32	$9.60

Danish himself (6 groups, ~250–286 hits/day): about **$0.09–$0.16 / day** → **~$3–$5 / month** OpenAI.

5. Planning takeaway

OpenAI normalize cost is **tiny** next to the Safepay plan (**PKR 7,000 / month**). The binding variable for AI spend is how many **new unique listing messages** arrive — not retries, and not duplicate reposts (those skip the API).

For multi-tenant forecasts: **monthly_openai_usd ≈ users × avg_groups × X × 0.0004 × 30**.

Source: Contabo prod DB · user danish@gmail.com (id 4) · openai_api_hits + whatsapp_messages / normalized_messages · Nov 2026.
Token costs estimated from prompt size + message length; OpenAI does not store token usage in our hit log yet.